

Supplementary Information

Resonant versus geometric anisotropy of the Purcell enhancement near a gold nanorod

S1. Drude–Lorentz representation of Johnson–Christy gold

For the time-domain body-of-revolution solver the Johnson–Christy permittivity [1] is represented as

$$\varepsilon(f) = \varepsilon_\infty - \frac{\sigma_0 f_0^2}{f^2 + if\gamma_0} + \sum_{j=1}^2 \frac{\sigma_j f_j^2}{f_j^2 - f^2 - if\gamma_j}, \quad (\text{S1})$$

with $f = 1/\lambda$ in μm^{-1} (Meep convention, $e^{-i\omega t}$). A least-squares fit over 440 nm to 1000 nm, weighted by $|\varepsilon|^{-1/2}$, gives the parameters of Table S1; the root-mean-square relative error is 1.6 % and the maximum 3.4 %. The same model is used in the Mie reference calculations, so that benchmark differences reflect discretisation only.

Table S1: Fit parameters (hcf gives energies in eV with $hc = 1.2398 \text{ eV } \mu\text{m}$).

term	f_j (μm^{-1})	hcf_j (eV)	γ_j (μm^{-1})	σ_j
ε_∞		2.229		
Drude	8.276	10.26	0.0490	0.701
Lorentz 1	2.208	2.737	0.509	1.156
Lorentz 2	2.755	3.416	0.010	2.101

S2. Three-dimensional FDTD set-up

Lumerical FDTD 2024 R1, cubic domain of side 600 nm with 12 PML layers, maximum simulation time 300 fs, auto-shutoff 10^{-5} (reached after 19 fs to 28 fs for cases B, C, D and 52 fs to 80 fs for the axial rod cases). A uniform override mesh ($\Delta = 2 \text{ nm}$ unless stated) covers the particle, gap and source; global mesh accuracy 3. Radiated power is the flux through six planar monitors on a 400 nm cube; 201 wavelengths between 500 and 900 nm. The dipole sits at $z = 35 \text{ nm}$, between two nodes of the 2 nm mesh, whereas in the free-space reference (case B) it sits at $z = 0$ on a node. The script that builds and runs all cases from scratch is provided with the data.

S3. Free-space reference and mesh sensitivity of the three-dimensional solver

Figure S1 shows the free-space check: T_B stays within $[0.993, 1.003]$ while $F_{p,B}$ oscillates between 0.93 and 1.13, which yields the unphysical $\eta_B = 1.08$ at 748 nm and bounds the normalisation uncertainty of F_p at $\pm 13 \%$ (3 % at 616 nm). Figure S2 shows the mesh sensitivity of case A: the radiative peak position (616(4) nm) and the efficiency at it are stable, whereas the peak heights depend on the sub-cell position of the source. Ratios between cases computed with the same mesh and source position are unaffected.

Table S2: All three-dimensional FDTD cases (raw definitions). η_{F_p} , η_T : efficiency at the Purcell and radiative peaks; $\eta_{\max}$: spectral maximum.

case	λ_{F_p}	$F_p^{\max}$	λ_T	$T_{\max}$	η_{F_p}	η_T	$\eta_{\max}$
A rod, axial, 5 nm	610	1496	616	119	7.6%	8.6%	19%
C rod, transverse	508	239	500	0.48	0.18%	0.21%	5.7%
D sphere, radial	508	659	532	6.03	0.76%	1.1%	12%
D $_{\perp}$ sphere, tangential	510	268	500	0.45	0.15%	0.18%	1.9%
D Mie (exact)	506	772	530	6.58	0.67%	—	11%
G3 gap 3 nm, 1-nm mesh	516	7198	614	252	0.05%	6.4%	7.7%
G10 gap 10 nm	610	553	614	46.8	8.1%	8.9%	31%
G20 gap 20 nm	610	91	618	9.9	9.6%	12%	74%
M3 mesh 3 nm	644	2027	648	184	8.7%	9.3%	18%
M1.5 mesh 1.5 nm	540	3752	620	165	0.05%	7.0%	9.6%

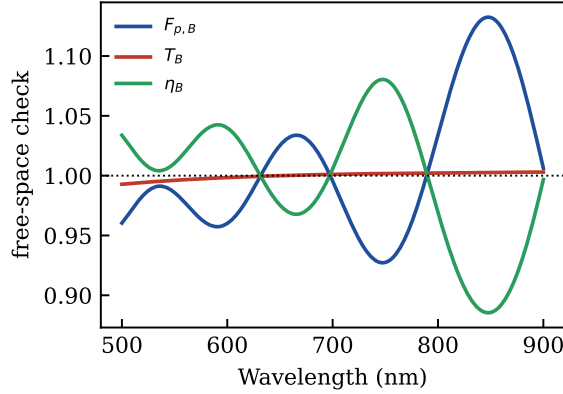


Figure S1: Free-space reference B of the three-dimensional solver.

S4. Body-of-revolution solver: method and benchmarks

Cylindrical-coordinate FDTD (Meep [2]) with field dependence $e^{im\varphi}$; axial dipole: E_z source, $m = 0$; transverse dipole: E_r source, $m = 1$. T is the ratio of the fluxes through a closed surface (top and bottom discs and a cylindrical side wall) around the particle with and without it. $F_p = (P_{\text{rad}} + P_{\text{abs}})/P_0$, where P_{abs} is the net inflow through a second closed surface that encloses only the particle, its top disc lying at mid-gap. The local density of states at the source cell gives the same order of magnitude but is spiky near the metal and is used only as a cross-check. The band 490 nm to 900 nm is assembled from three runs with pulses covering 460 to 610, 540 to 760 and 630 nm to 990 nm, of which only 490 to 570, 570 to 700 and 700 nm to 900 nm are kept: a broadband single run produced errors up to 37 % at the band edges, where the source spectrum is weak. Run time 30 $\mu\text{m}/c$ after the source (identical F_p to 60); PML thickness 0.15 μm , with a 0.3 μm check at the finest grid. To avoid long-lived numerical ringing, the Lorentz-pole damping in the gold fit is bounded below by 0.2 μm^{-1} (same fit accuracy).

Benchmark against the exact Mie solution for a dielectric sphere ($\varepsilon = 4$, $R = 15.874$ nm, radial dipole at 5 nm): a grid of 2 nm reproduces F_p within 1 % in the band centres; a 4 nm grid gives a uniform 4 % to 5 % overestimate with a maximum deviation of 12 %. For gold (radial dipole, 5 nm gap) at a 2 nm grid, T deviates from Mie by about 10 % while the power-balance F_p is about three times too large; at a 20 nm gap the F_p error drops to 24 % and T agrees within 5 %. The F_p error is therefore a near-field discretisation error of the staircased metal surface. Grids of 2, 1.33 and 1 nm with linear extrapolation to zero spacing are summarised in Table S3.

PML thickness 0.15 versus 0.3 μm at the 1 nm grid changes T by 0.1 % (median) but F_p by 9 %.

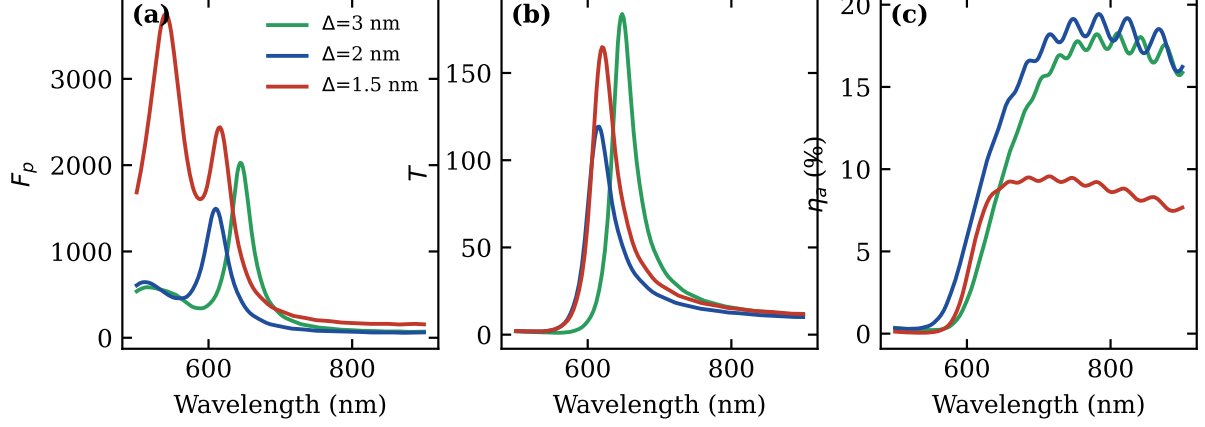


Figure S2: Mesh sensitivity of case A in the three-dimensional solver: (a) F_p , (b) T , (c) η_a for override meshes of 3, 2 and 1.5 nm.

Table S3: Body-of-revolution solver against exact Mie theory (gold sphere, radial dipole, 5 nm gap) and grid convergence of the nanorod (axial dipole, 5 nm gap). Errors are medians over 490 nm to 900 nm.

grid (nm)	sphere: $ \Delta F_p $	sphere: $ \Delta T $	rod: $T_{\max} (\lambda_T)$	rod: NA 0.9 fraction
2.00	215%	10.2%	149 (631 nm)	0.181
1.33	108%	8.4%	134 (617 nm)	0.181
1.00	93%	7.2%	125 (611 nm)	0.180
extrapolated	48%	4.4%	—	—

The radiative quantities are therefore converged to the level quoted in the main text, whereas F_p in a 5 nm gap is not converged at grids of 1 nm with a staircased surface and is not used. The transverse ($m = \pm 1$) configuration with a single E_r source on the axis gave unphysical (negative) total rates and is likewise not used; transverse results are taken from the three-dimensional solver. All runs were executed on GitHub-hosted runners (workflow `.github/workflows/meep-runs.yml`); raw outputs are in `isi/results/ci`.

References

- [1] P. B. Johnson and R. W. Christy, Phys. Rev. B **6**, 4370 (1972).
- [2] A. F. Oskooi *et al.*, Comput. Phys. Commun. **181**, 687 (2010).